

WindForge Technical Specification

Version: Opus 4.7 audit pass, May 2026 **Status:** All 702 tests passing (678 core + 24 UI) **Audience:** Engineers, AI agents, contributors needing a single document that explains *what WindForge is, what it measures, every file, and where to extend it*.

1. Executive summary

WindForge is a TypeScript decision-support platform for wind turbine site suitability. It is **not** an "AI placement" tool. It pulls from public meteorological, terrain, and infrastructure datasets, runs a stack of physics-grounded models (wind shear, Jensen / Bastankhah wakes, ISO 9613-2 noise, solar ephemeris shadow flicker, Gumbel extreme wind, Weibull energy yield, IEC 61400-1 turbulence, financial NPV/IRR/LCOE) and returns explained, weighted scores plus per-domain numerical results. Every output is inspectable; humans always make the final call.

Architecture: pnpm + Turborepo monorepo with a strict split between a pure-TS `core` (no React, no DOM) and a React `ui` layer.

```
packages/  
  core/   pure TypeScript, runs in Node / Deno / Bun / browser  
  ui/     React 18, Recharts, react-leaflet  
  demo/   Next.js 15 showcase app
```

1.1 Bug fixes & hardening applied in this pass

#	File	Defect	Fix
1	packages/ui/src/components/WindSpeedDistribution.tsx	useMemo was placed after an early if (loading) return ..., violating React's Rules of Hooks. The component would crash when loading toggled.	Hook moved above all conditional returns; data?.bins? made null-safe.
2	packages/ui/src/components/SiteMap.tsx (BoundsWatcher)	Debounce setTimeout was never cleared on unmount, retaining a closure over the Leaflet map after teardown (memory leak + potential callback against unmounted map).	Added useEffect cleanup that clears the timer on unmount.
3	packages/core/src/cumulative/cumulative-impact.ts	Existing turbine IDs were offset by a hard-coded +10000, which silently collides if proposed turbine IDs are >= 10000 (e.g.	Offset is now max(proposedId) + 1, computed dynamically, then i per turbine -

		OSM-derived numeric IDs are routinely larger).	guaranteed unique.
4	packages/core/src/datasources/osm-overpass.ts and packages/core/src/constraints/constraint-queries.ts	Voltage filter <code>^[1-9][0-9]{5,}\$</code> skipped OSM tags storing semicolon-separated voltages such as "132000;33000", materially undercounting transmission lines on combined-circuit pylons.	Regex relaxed to <code>`(^</code>

All fixes are covered by the existing test suite (no regressions). No production behaviour changed for callers using clean inputs - the fixes only correct edge cases that were silently wrong.

2. What it measures (the data flow)

+-----+	+-----+	+-----+
PUBLIC APIs	-> CORE ENGINE	-> OUTPUT SURFACES
+-----+	+-----+	+-----+
NASA POWER	wind shear (a)	SiteAnalysis
Open-Elevation	6-factor scoring	SiteAssessment
OSM Overpass	constraint det.	EnergyYieldResult
OSM Nominatim	wake (Jensen/	NoiseResult +
ERA5 (opt.)	Bastankhah)	ETSU compliance
CERRA (opt.)	noise ISO 9613-2	ShadowFlicker +
Met-mast CSV	shadow flicker	compliance
+-----+	Weibull AEP	IecSiteReport
	Gumbel extremes	FinancialResult
	turbulence (IEC)	TurbineLayout
	terrain speed-up	ConstraintReport
	viewshed (ZTV)	CumulativeImpact
	financial NPV	(all stream
	+-----+	through React UI
		+ PDF export)
		+-----+

Scoring factors (composite 0-100)

Factor	Default weight	Source	Score driver
Wind resource	0.35	NASA POWER	Hub-height speed (extrapolated), CV, directional consistency
Terrain suitability	0.20	Open-Elevation	Slope %, elevation, roughness class

Grid proximity	0.15	OSM Overpass	Distance to nearest substation (60%) and 100 kV+ line (40%)
Land-use compatibility	0.15	OSM Overpass	Hard exclusions, soft setbacks, positive indicators
Planning feasibility	0.10	Nominatim + OSM	Country framework, nearby precedents, density proxy
Access logistics	0.05	OSM Overpass	Best road category within 5 km

Hard constraints (any factor < 20) are surfaced regardless of composite score. Weights normalise to 1.0 automatically.

Wind-shear extrapolation

Power-law profile is applied with reference height `h_ref` = 50 m when NASA POWER `WS50M` is non-zero, otherwise 2 m:

$$v_{hub} = v_{ref} \left(\frac{h_{hub}}{h_{ref}} \right)^{\alpha}$$

`alpha` is derived from terrain roughness class (0.10 offshore -> 0.35 urban), not hardcoded.

3. File-by-file reference

Every TypeScript file under `packages/core/src` and `packages/ui/src` . One line summary, then key exports.

3.1 `packages/core/src` - root

File	Role
<code>index.ts</code>	Public API surface: re-exports types and functions. Keeps the package's import boundary explicit.
<code>cli.ts</code>	Command-line analyser. <code>npx tsx packages/core/src/cli.ts <lat> <lng> [--hub-height N]</code> . Pretty-prints score, factor breakdown, hard constraints, and source health.

3.2 `analysis/` - derived statistics from raw wind history

File	What it does
<code>wind-analysis.ts</code>	Pure functions: <code>computeWindTrend</code> (linear regression + R^2), <code>computeSeasonalHeatmap</code> (month x hour grid), <code>computeMonthlyBoxPlots</code> (Tukey IQR + outliers), <code>computeDiurnalProfile</code> , <code>computeSpeedDistribution</code> (Weibull k/c MoM fit), <code>computeYearOverYear</code> .
<code>turbulence.ts</code>	IEC 61400-1 turbulence intensity: bins by wind speed, computes mean and representative TI at 15 m/s, classifies A/B/C/exceeds_A.
<code>extreme-wind.ts</code>	Gumbel Type-I fit (<code>fitGumbel</code> , <code>gumbelQuantile</code>) for 1-, 50-, 100-year return periods; assigns IEC wind class I/II/III/S.
<code>data-quality.ts</code>	Scores wind history completeness, gap length, and recency.
<code>mcp-analysis.ts</code>	Measure-Correlate-Predict: linear regression between short on-site mast and long-term reanalysis.

index.ts	Barrel export for the analysis subpackage.
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3.3 cache/ - request and tile caches

File	What it does
index.ts	Generic in-memory TTL cache used by every data-source client.
spatial-cache.ts	Geohash-keyed tile cache for elevation grids and Overpass responses to avoid duplicate fetches when sampling a polygon.

3.4 constraints/ - hard/soft/info constraint engine

File	What it does
constraint-definitions.ts	19 ConstraintDefinition records across environmental, aviation, military, heritage, residential, infrastructure, water, terrain. Each with severity, default setback metres, and human description.
constraint-queries.ts	Single batched Overpass query builder for a site's bounding box (expanded by max setback). 24 h cache, 30 s timeout, 1 retry after 5 s.
constraint-detector.ts	Walks Overpass elements, classifies each, computes distance to site polygon edge, builds SiteConstraintReport with hard / soft / info lists, nearest-receptor table, exclusion zones, and recommendation enum.
exclusion-geometry.ts	Buffers each hard constraint by its setback, intersects with site polygon, returns ExclusionZone[] with overlap area.
index.ts	Barrel export.

3.5 cumulative/ - combined existing + proposed impact

File	What it does
cumulative-impact.ts	assessCumulativeImpact(proposed, existing, receptors, options) -> noise (proposed only and combined), shadow flicker, optional viewshed. Now uses dynamic ID offset to prevent collisions.
index.ts	Barrel export.

3.6 datasources/ - external API clients

All clients implement: retry with exponential backoff, abort signal support, in-memory TTL cache, structured `Result<T, ScoringError>` returns.

File	API	What it returns
nasa-power.ts	NASA POWER (RE community)	WindDataSummary for fast scoring; MonthlyWindHistory (1981-present), DailyWindData, HourlyWindData for charts. Multi-height: WS2M / WS10M / WS50M / WD10M / WD50M.

open-elevation.ts	Open-Elevation (SRTM 30 m)	ElevationData: elevation, slope %, aspect, roughness class. Samples a 9-point grid and fits gradient.
osm-overpass.ts	OSM Overpass API	Grid infrastructure (lines ≥ 100 kV via GRID_VOLTAGE_REGEX, substations), land use, road access, nearby wind farms. Each cached 24 h.
nominatim.ts	OSM Nominatim	Reverse geocoding for country/region context. 1 req/s rate limited, mandatory User-Agent.
era5.ts	ECMWF ERA5 (optional)	High-resolution (~ 31 km) reanalysis for higher-confidence wind data.
cerra.ts	Copernicus CERRA (optional)	European 5.5 km reanalysis.
met-mast-parser.ts	Local CSV	Parses on-site mast files for MCP correlation.
index.ts	Barrel export.	

3.7 scoring/ - the 6-factor engine

File	What it does
engine.ts	analyseSite(options). Validates coordinate, normalises weights, fires all data sources in parallel via Promise.allSettled, hands each result to its factor scorer, assembles SiteAnalysis. Hard constraints (< 20) escalate. Records sources used / failed in metadata.
wind-resource.ts	Combines hub-height speed (60%), consistency CV (25%), directional stability (15%). Confidence from dataYears.
terrain-suitability.ts	Slope (50%), elevation (25%), roughness (25%). Slope $\geq 30\%$ scores 0.
grid-proximity.ts	Substations weighted 60%, lines 40%. Distance bands 5/15/30/50 km.
land-use.ts	Hard constraints score 0 with constraint[] raised. Soft de-duplicated by type so multiple residential polygons aren't penalised separately. Positive indicators boost.
planning.ts	Country list, precedent (existing wind farms within 20 km), density proxy. Always low/medium confidence; explicitly disclaims formal assessment.
access.ts	Best road category and distance. Primary ≤ 2 km $\rightarrow$ 80-100; no roads $\rightarrow$ hard constraint.
index.ts	Barrel + DEFAULT_WEIGHTS and normaliseWeights.

3.8 wake/ - directional wake-loss models

File	What it does
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jensen-wake.ts	Jensen (1983) top-hat wake. Includes thrust-curve generation from power curve via actuator-disc theory and Newton solve of $C_p = 4a(1-a)^2$. Wake-rotor circle overlap by analytical area intersection. RSS / Katic superposition.
bastankhah-wake.ts	Gaussian wake (Bastankhah & Porte-Agel, 2014) with $\sigma_y = k^* x + \epsilon d$ near-wake correction.
wake-loss-calculator.ts	<code>calculateDirectionalWakeLoss(layout, turbine, windData, model)</code> integrates the 36-sector wind rose, runs the chosen wake model per sector, weights by sector frequency, returns sector + total wake loss %.
index.ts	Barrel.

3.9 energy/ - AEP, layout

File	What it does
aep-calculator.ts	Weibull-based annual energy production with power-curve integration, air-density correction (lapse rate from elevation), 7-item loss stack (wake, electrical, availability, environmental, icing, hysteresis, curtailment), P50 / P75 / P90 ($z = 0 / 0.674 / 1.282$ on log-normal uncertainty). 12-month seasonal split.
layout-optimiser.ts	Greedy hill-climb layout. Respects spacing rules (3D crosswind, 5D downwind), site polygon, exclusion zones. Optimises directional AEP through wake model.
turbine-layout.ts	Helpers: grid layout, edge layout, validate spacing.
index.ts	Barrel.

3.10 financial/ - economics

File	What it does
financial-model.ts	LCOE, NPV, simple + discounted payback, IRR via Newton-Raphson. 25-year cashflow with degradation, opex, capex, decommissioning.
scenario-analysis.ts	Sensitivity analysis: tornado diagram-style deltas for capex / opex / wind / price.
index.ts	Barrel.

3.11 noise/ - acoustics

File	What it does
noise-propagation.ts	ISO 9613-2 implementation: geometric divergence, atmospheric absorption (frequency-dependent table), ground effect (G-factor), barrier screening.
etsu-assessment.ts	UK ETSU-R-97 compliance: 35-43 dB(A) day, 43 dB(A) night, plus 5 dB amenity above background; relaxed to 45 dB at financially-involved receptors.
noise-contours.ts	Generates iso-noise contour polygons by sampling a grid then marching-squares.

index.ts	Barrel.
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3.12 shadow/ - flicker

File	What it does
solar-position.ts	Meeus astronomical algorithm: solar azimuth + elevation per UTC datetime + lat/lng.
shadow-flicker.ts	Hour-by-hour annual simulation. For each hour with sun above 1°, checks each turbine's shadow geometry vs each receptor (rotor angular width, shadow length, distance cap). Returns hours/year per receptor.
shadow-calendar.ts	Month x hour matrix of flicker days for mitigation scheduling (e.g. shutdown windows).
index.ts	Barrel.

3.13 terrain/ - topographic effects

File	What it does
elevation-grid.ts	Builds dense ElevationGrid from Open-Elevation tiles.
terrain-speedup.ts	Jackson-Hunt linearised speed-up over hills. Identifies ridge orientation, returns speed-up factor and effective wind multiplier.
rix-calculator.ts	RIX (Ruggedness Index): % of terrain exceeding 30% slope within a radius. RIX > 5% indicates non-trivial flow distortion; > 30% triggers warnings.
index.ts	Barrel.

3.14 turbines/ - model library

File	What it does
turbine-library.ts	12+ built-in models from 660 kW (Vestas V47) to 8 MW (Siemens Gamesa SG 8.0-167) with rated power, rotor diameter, hub-height options, full power curves, sound power level.
power-curve-parser.ts	CSV import for custom turbines; validates monotonic wind speed, plausible Cp.
index.ts	Barrel + getAllTurbines, getTurbineById.

3.15 site/ - polygon-based assessment

File	What it does
site-boundary.ts	createBoundary(polygon, name?), parseBoundaryFromGeoJSON, parseBoundaryFromKML. Computes centroid, area (spherical excess), bounding box.
site-assessment.ts	Top-level assessSite(boundary, options): samples grid points inside polygon, scores each, aggregates factors, runs constraint detector, optionally calculates AEP for a chosen turbine.

index.ts	Barrel.
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3.16 visual/ - viewshed

File	What it does
viewshed.ts	Zone of Theoretical Visibility. Bilinear elevation sampling + line-of-sight checks with earth-curvature drop $d^2 / (2R)$.

3.17 reporting/ - structured outputs

File	What it does
iec-report.ts	Builds IecSiteReport aligned with IEC 61400-1 Ed.4 + 61400-12-1 Ed.2: wind conditions, turbulence class, extreme wind class, energy summary, suitability recommendation. Pure data structure (rendered by UI / PDF).

3.18 validation/

File	What it does
data-validator.ts	Cross-checks: speed range plausibility, missing-month detection, direction sanity, height consistency.

3.19 utils/

File	What it does
geo.ts	Haversine distanceKm, clamp, linearScale, isValidCoordinate.
geometry.ts	Polygon area (spherical excess), point-in-polygon (ray casting), edge distance, circle buffer, polygon overlap area.
wind-shear.ts	extrapolateWindSpeed, roughnessClassToAlpha, REFERENCE_HEIGHT_M.
cache.ts	Re-export of cache/index.ts.
fetch.ts	Shared retry + abort wrapper around fetch.
elevation-profile.ts	Samples elevation along a great-circle arc.

3.20 types/ - shared interfaces and discriminated unions

analysis.ts , constraints.ts , datasources.ts , energy.ts , errors.ts , financial.ts , met-mast.ts , noise.ts , result.ts , shadow.ts , site.ts , terrain.ts , turbines.ts , wake.ts , wind-assessment.ts . All branded; Result<T, E> is the canonical error channel.

4. UI components (packages/ui/src)

All components: named-export, accept className and theme , render loading & error states, no direct API calls (data flows through core via hooks).

Component	Purpose
WindSiteScorer.tsx	Drop-in widget: map + coord inputs + score card + sliders + export.
SiteMap.tsx	Leaflet map with click-to-analyse, animated loading pin, optional heatmap overlay, debounced bounds-change emitter. Memory-leak in BoundsWatcher fixed in this pass.
ScoreCard.tsx	Composite score circle, hard-constraint banner, per-factor bars, optional Wind Rose.
WeightSliders.tsx	Six sliders, live re-normalisation, sum indicator.
WindRose.tsx	16-point compass radar built on Recharts. Helpers: degreesToCompass, emptyRoseData, DEFAULT_WIND_BANDS.
WindTrendChart.tsx	Monthly speed line + linear-regression trend overlay; shows slope per year and R^2 .
SeasonalHeatmap.tsx	SVG month x hour grid, blue->green->red colour scale, integrated legend.
MonthlyBoxPlot.tsx	Recharts ComposedChart with custom box & whisker (IQR bar + ErrorBar whiskers + median/mean lines).
DiurnalProfile.tsx	Hourly area chart with min/max band and mean line.
WindSpeedDistribution.tsx	Histogram + Weibull curve overlay. Hooks-order bug fixed in this pass.
ScenarioCompare.tsx	Up to 4 sites side-by-side; highlights best per factor and overall winner.
SiteBoundaryEditor.tsx	Click-to-add polygon vertices, manual coord entry, GeoJSON / KML upload.
ConstraintPanel.tsx	Severity-coloured constraint groups, recommendation badge, nearest-receptor table.
ConstraintMap.tsx	SVG aspect-corrected mini-map of boundary, exclusion zones, clustered constraint pins.
EnergyYieldCard.tsx	Gross/net AEP, capacity factor, P50/P75/P90, monthly production, assumptions.
LossStackChart.tsx	Horizontal stacked bar of the 7-item loss stack.
TurbineSelector.tsx	Grouped picker by power class.
SiteAssessmentView.tsx	Top-level layout for a full polygon assessment.
ExportButton.tsx	Lazy-loads jsPDF + html2canvas; renders a structured PDF including chart screenshots.

Hooks

Hook	Role
use-wind-data.ts	Imperative <code>fetch(coordinate)</code> returning <code>WindDataSummary</code> .
use-site-score.ts	<code>analyse(options)</code> with abort-controller-backed cancellation; ignores stale results when superseded.

use-map- interaction.ts	Selected coord + animated map pin state.
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Styles

styles/theme.ts defines the WindSiteTheme token set; all components consume CSS custom properties (--wsi-primary , --wsi-text , etc.) so consumers can override without rebuilding.

5. Test inventory (38 core suites, 1 UI suite, 702 total)

tests/access.test.ts	16		tests/met-mast.test.ts	20
tests/aep-calculator.test.ts	30		tests/nasa-power-extended.test.ts	12
tests/cache.test.ts	6		tests/noise-model.test.ts	49
tests/cerra.test.ts	9		tests/nominatim.test.ts	9
tests/constraint-definitions.test.ts	9		tests/osm-overpass.test.ts	25
tests/constraint-detector.test.ts	12		tests/planning.test.ts	18
tests/cumulative-impact.test.ts	7		tests/result.test.ts	3
tests/data-validator.test.ts	19		tests/shadow-flicker.test.ts	39
tests/engine.test.ts	11		tests/site-boundary.test.ts	13
tests/era5.test.ts	8		tests/spatial-cache.test.ts	12
tests/financial-model.test.ts	26		tests/terrain-flow.test.ts	18
tests/geometry.test.ts	31		tests/terrain-suitability.test.ts	9
tests/grid-proximity.test.ts	15		tests/turbine-layout.test.ts	8
tests/iec-report.test.ts	8		tests/turbine-library.test.ts	16
tests/land-use.test.ts	23		tests/utils.test.ts	21
tests/layout-optimiser.test.ts	12		tests/viewshed.test.ts	11
tests/wake-model.test.ts	46		tests/wind-analysis.test.ts	36
tests/wind-assessment.test.ts	23		tests/wind-resource.test.ts	11
tests/wind-rose.test.ts	25		tests/wind-shear.test.ts	12
ui/tests/chart-components.test.ts	24			
total = 702				

6. Public API consumption modes

6.1 Drop-in widget

```
import { WindSiteScorer } from '@jamieblair/windforge';

<WindSiteScorer defaultCenter={{ lat: 55.86, lng: -4.25 }} hubHeightM={100} />
```

6.2 Headless SDK

```
import { analyseSite, calculateAep, getTurbineById } from '@jamieblair/windforge-core';

const a = await analyseSite({ coordinate: { lat: 55.86, lng: -4.25 }, hubHeightM: 100 });
const aep = calculateAep(windData, getTurbineById('vestas-v110-2000'))!;
```

6.3 Site (polygon) assessment

```
import { createBoundary, assessSite } from '@jamieblair/windforge-core';
const boundary = createBoundary([...polygon], 'My Site');
const assessment = await assessSite(boundary, { hubHeightM: 100 });
```

6.4 CLI

```
npx tsx packages/core/src/cli.ts 55.86 -4.25 --hub-height 100
```

7. Architectural invariants (preserved across this pass)

- Core has zero dependencies on React, DOM, or Node-only built-ins beyond `fetch` and `setTimeout`.
- UI never calls a remote API directly; all data flows through hooks that go through core.
- Errors travel via `Result<T, ScoringError>`; thrown exceptions only for programmer errors.
- No `any`, no default exports, no barrel re-exports beyond package entry points.
- Numbers carry units in their variable names (`distanceKm`, `speedMs`, `elevationM`, `hubHeightM`).
- Every scoring function returns a `confidence: 'high' | 'medium' | 'low'` plus a human `detail` string.
- `Promise.allSettled` for fan-out so any single failed source degrades to neutral with low confidence rather than killing the analysis.

8. Suggested next extensions (for the next AI to work on)

These are gaps observed during the audit, ranked by leverage.

1. **Bias-correct NASA POWER against ERA5 / CERRA when both are present.** The clients exist; a small reconciliation layer in `analysis/` would lift wind-resource confidence to `high` for European sites.
2. **Cumulative wake from existing wind farms.** `cumulative-impact.ts` already merges turbine lists for noise + flicker; extend to `calculateDirectionalWakeLoss` so AEP estimates account for upstream existing wakes.
3. **Tile-based SiteMap heatmap pre-fetch.** The `BoundsWatcher` now emits bounds reliably (post-fix); a worker-side coarse-grid scorer could populate heatmap points without blocking interactions.
4. **MCP server wrapper (`packages/mcp/`).** Expose `analyseSite`, `assessSite`, `fetchMonthlyWindHistory`, `calculateAep` as MCP tools so any LLM can call them. Stdio transport, no auth.
5. **Persistent cache adapter.** Current `cache/index.ts` is in-memory only. An optional IndexedDB / SQLite adapter would survive page reloads and CLI invocations.
6. **Offshore-mode terrain scoring.** `roughnessClass=0` already returns 85; add bathymetry source (GEBCO) and update `terrain-suitability.ts` to score water depth instead of slope.
7. **Sector management for noise / shadow.** Where ETSU or shadow limits would otherwise fail, suggest curtailment sector schedules using `shadow-calendar.ts` matrices and noise direction-of-arrival.
8. **Battery + grid-firming module under `financial/`.** Pair AEP P50/P90 spread with optional storage sizing for capacity-firming LCOE.

9. Glossary (selected)

Term	Meaning
AEP	Annual Energy Production (MWh/year).
Capacity factor	AEP / (rated power x 8760 h).
Cp / Ct	Power coefficient / thrust coefficient.
ETSU-R-97	UK noise guidance for wind farms.
Gumbel	Type-I extreme value distribution used for return-period winds.
IEC 61400	International standard family for wind turbine generator systems.
LCOE	Levelised cost of energy.
MCP	Measure-Correlate-Predict.
P50 / P75 / P90	AEP exceedance probabilities (50%, 75%, 90%).
RIX	Ruggedness index (% of area > 30% slope).
Weibull	Two-parameter speed distribution (k, c).
ZTV	Zone of Theoretical Visibility.

End of specification. Generated for use as a single-document briefing for AI agents and engineers extending WindForge.